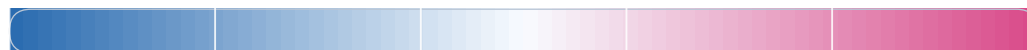


Compartment-aware RNA-CyTOF immune-state decoupling map

A



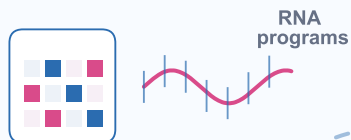
coordinated
immune response

PC1 variance 55.9%

221/499 cases

decoupled inflammatory /
MHC-II-low state

B Whole-blood RNA layer



**Inflammatory / S100 /
host-response** ↑

S100A8 / S100A12 / HK3

**MHC-II / CD74 /
HLA-DR** ↓

CD74 / CIITA / HLA-DRA

Decoupling index =
 $z(\text{HR}) - z(\text{MHC-II/CD74})$

7 cohorts; median $\rho \approx -0.60$

C Paired RNA-CyTOF bridge

COMBAT sepsis-only
40 rows / 34 participants



paired association

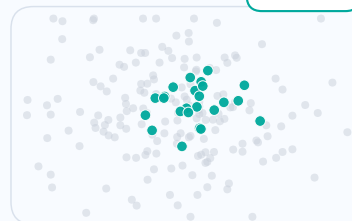
HLA-DR core $\rho = -0.559$

MHC-II/CD74 $\rho = -0.517$

Decoupling index $\rho = 0.462$

D CyTOF event-derived layer

**Residual CD3/CD14
coevent signal**



Activation-state context



378-pair marker-null:
pair specificity unresolved



Claim boundary

Association-only

Artifact-aware coevent summary

Pair specificity unresolved

Prospective validation
required